Sebastian Adolfo Cuesta Hoyos

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RESEARCH PROFILE

Multidisciplinary chemist with a PhD in Chemical Biology, 10+ years of academic research experience, and 5+ years in pharmaceutical R&D. My work combines enzyme discovery, biocatalysis, and data-driven drug discovery, integrating computational and experimental approaches to investigate enzyme function, molecular mechanisms, and therapeutic biosynthesis. I have led collaborative, cross-disciplinary projects, authored over 30 peer-reviewed publications, and will begin a Marie Skłodowska-Curie Actions Postdoctoral Fellowship in summer 2027.

SELECTED FELLOWSHIPS & AWARDS

- Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship — Awarded, European Commission / Horizon Europe, 2026; to begin summer 2027.
- Dean's Doctoral Scholarship Award, University of Manchester, 2021
- SENESCYT Scholarship for MSc studies, Ecuador, 2015
- Best Graduate in Chemistry, Pontifical Catholic University of Ecuador, 2013
- Academic & Sport Award, Pontifical Catholic University of Ecuador, 2009

RESEARCH INTERESTS

Enzyme discovery and biocatalysis; chemical biology; molecular mechanisms and chemical reactivity; computational and data-driven drug discovery; molecular docking and molecular dynamics; machine learning in chemistry and life sciences; antimicrobial peptides and anti-infective therapeutics.

EDUCATION

PhD in Chemical Biology - **University of Manchester** | *UK* | *2025*

- Thesis: Discovery, Characterisation, and Evolution of New Enzymes for the Biosynthesis of Therapeutics

MSc in Drug Discovery and Translational Biology - **University of Edinburgh** | *UK* | *2016*

- Thesis: Computer-Aided Design of Cks1-Skp2 Staple Peptide Inhibitors with Chemotherapeutic Potential

BSc in Chemistry, First Class Honours - **Pontifical Catholic University of Ecuador** | *Quito, Ecuador* | *2014*

- Thesis: Chemical, computational, and pharmacological study of ibuprofen

RESEARCH & PROFESSIONAL EXPERIENCE

Research Associate in Biochemistry / Chemical Biology - **The University of Manchester** | *UK* | *2025–Present*

- Investigate *Mycobacterium tuberculosis* nutrient acquisition strategies to identify new biological pathways and potential therapeutic targets.
- Study mycobacterial membrane protein function, substrate recognition, and biochemical mechanisms.
- Train and support PhD and MSc students in experimental design, data interpretation, and laboratory workflows.

Researcher - Theoretical & Computational Chemistry Lab - **San Francisco University of Quito** | *Ecuador* | *2017–2026*

- Apply machine learning, molecular docking, molecular dynamics, and quantum chemistry to drug repurposing, molecular modelling, and mechanistic studies.
- Collaborate with researchers across chemistry, biology, and pharmaceutical sciences to develop computational workflows for therapeutic discovery.

Graduate Teaching Assistant - **University of Manchester** | *UK* | *2023–2025*

- Assisted first-year undergraduate chemistry laboratory teaching, including supervision, experimental guidance, and student support.

Researcher - Computational Chemistry Lab - **Pontifical Catholic University of Ecuador** | *Quito, Ecuador* | *2014–2022*

- Modelled reaction mechanisms, transition states, and electronic structures using quantum chemical calculations.
- Developed computational workflows to study structure-reactivity relationships and biologically relevant molecular systems.

R&D Analyst - Scale-up and Technology Transfer - **LIFE Laboratories** | *Ecuador* | *2019–2021*

- Led technology transfer and scale-up of pharmaceutical products from development to manufacturing environments.
- Conducted process development, validation activities, and regulatory documentation aligned with pharmaceutical standards.

Formulation Analyst - Research & Development - **LIFE Laboratories** | *Ecuador* | *2017–2019*

- Developed and reformulated pharmaceutical products, supporting formulation optimisation and analytical evaluation.
- Designed and validated analytical methods for QC and R&D applications.

Quality Control Analyst - LIFE Laboratories | *Ecuador* | *2013–2015*

- Performed quality control testing of pharmaceutical products following regulatory and laboratory standards.
- Prepared working standards from primary USP reference standards and supported analytical documentation.

SCIENTIFIC MENTORING & CAPACITY BUILDING

- Mentored undergraduate and postgraduate students in Ecuador in thesis development, research design, data analysis, scientific writing, and manuscript preparation.
- Supported early-career researchers in transforming thesis projects into publishable scientific manuscripts and research outputs.
- Provided guidance in computational chemistry, molecular modelling, docking, molecular dynamics, QSAR, and machine learning approaches applied to drug discovery and chemical biology.
- Contributed to scientific capacity building in Ecuador by helping students develop research independence, critical thinking, and publication-oriented academic skills.

TECHNICAL SKILLS

Computational chemistry and modelling: Quantum chemistry/QM (Gaussian), molecular dynamics (Gromacs), molecular docking (AutoDock Vina), protein modelling (AlphaFold, SWISS-MODEL), QSAR, mechanistic modelling

Machine learning and data analysis: Python, scikit-learn, WEKA, Mobydigs, data-driven model development and interpretation

Analytical and biochemical methods: HPLC, LC-MS, GC-MS, UV-Vis, NMR, IR, FPLC, enzymatic assays, protein purification

Molecular biology and biotechnology: Cloning, PCR, mutagenesis, protein expression, protein purification, X-ray crystallography, metagenomics

Pharmaceutical R&D: GxP, formulation development, organic synthesis, analytical method development, ICH and USP guidelines, scale-up and technology transfer

Software and tools: Microsoft Office, ChemDraw, OriginPro

PEER REVIEW & ACADEMIC SERVICE

Peer reviewer for journals including Structural Chemistry, Computers in Biology and Medicine, Journal of Biomolecular Structure and Dynamics, European Journal of Pharmaceutical Sciences, Journal of Molecular Modeling, and related journals.

SELECTED ORAL & POSTER PRESENTATIONS

- Enzymatic cascades for stereoselective and regioselective amide bond assembly, Biotrans, Basel, 2025 (poster)
- Peptides with beta-lactone warheads, Novel Enzyme Conference, Budapest, 2025 (oral)
- Using ML to predict ABS substrate scope, PGR Conference, University of Manchester, 2024 (oral)
- Enzymatic approaches to construct next-generation oligonucleotide therapeutics, PGR Conference, University of Manchester, 2023 (poster)
- Drug Discovery and Design, USFQ Seminar, Quito, 2017 (oral)

SELECTED PUBLICATIONS

Selected from peer-reviewed publications; full publication list available on Google Scholar, ORCID, and personal website.

1. Torri D.; Bering L.; Yates L.R.L.; Angiolini S.M.; Xu G.; Cuesta-Hoyos S.; Shepherd S.A.; Micklefield J., Enzymatic Cascades for Stereoselective and Regioselective Amide Bond Assembly, 2025, Angewandte Chemie International Edition, DOI: 10.1002/anie.202422185

2. Xu G.; Torri D.; Cuesta-Hoyos S.; Panda D.; Yates L.R.L.; Zallot R.; Bian K.; Jia D.; Iorgu A.I.; Levy C.; Shepherd S.A.; Micklefield J., Cryptic enzymatic assembly of peptides armed with β-lactone warheads, 2024, Nature Chemical Biology, DOI: 10.1038/s41589-024-01657-7

3. Cuesta S.A.; Moreno M.; López R.A.; Mora J.R.; Paz J.L.; Márquez E.A., ElectroPredictor: An Application to Predict Mayr's Electrophilicity E through Implementation of an Ensemble Model Based on Machine Learning Algorithms, 2023, Journal of Chemical Information and Modeling, DOI: 10.1021/acs.jcim.2c01367

4. Valencia J.; Rubio V.; Puerto G.; Vasquez L.; Bernal A.; Mora J.R.; Cuesta S.A.; Paz J.L.; Insuasty B.; Abonia R.; Quiroga J.; Insuasty A.; Coneo A.; Vidal O.; Márquez E.; Insuasty D., QSAR Studies, Molecular Docking, Molecular Dynamics, Synthesis, and Biological Evaluation of Novel Quinolinone-Based Thiosemicarbazones against *Mycobacterium tuberculosis*, 2023, Antibiotics, DOI: 10.3390/antibiotics12010061

5. Cabrera N.; Cuesta S.A.; Mora J.R.; Calle L.; Márquez E.A.; Kaunas R.; Paz J.L., In Silico Searching for Alternative Lead Compounds to Treat Type 2 Diabetes through a QSAR and Molecular Dynamics Study, 2022, Pharmaceutics, DOI: 10.3390/pharmaceutics14020232

6. Cuesta S.A.; Reinoso C.; Morales F.; Pilaquinga F.; Morán-Marcillo G.; Proaño-Bolaños C.; Blasco-Zúñiga A.; Rivera M.; Meneses L., Novel antimicrobial cruzioseptin peptides extracted from the splendid leaf frog, *Cruziohyla calcarifer*, 2021, Amino Acids, DOI: 10.1007/s00726-021-02986-w

LANGUAGES

Spanish: Native

English: Fluent